Supplementary File S2: Agent-Mediated Thermal-Twin Integration Test

This file summarizes one supervised agent-to-device test reported in the manuscript. The source record contains 21 passing assertions: three simulator and device setup checks, six platform and protocol checks, one channel and binding check, two agent-context and acquisition checks, five write and readback checks, two process and acquisition-response checks, and two agent refetch checks. The summary preserves the order and ownership of the recorded actions; it is not a complete runtime replay package.

1 Configuration

An isolated platform instance connected one acquisition node and three setpoint nodes to a separate gray-box extrusion/casting thermal simulator over Modbus TCP. The simulator used first-order thermal inertia and delay and was not calibrated against a physical production line. An independent Modbus client read the simulator's registers, separately from platform readback. One OMP harness with a local Qwen model operated in a manual-approval channel.

2 Recorded Sequence

Table 1: Distinct actions and their associated device and process evidence.

Stage	Action and device-side record	Process and acquisition record
Context retrieval	The agent queried its bound nodes, the active line context, and the acquisition service. One initial query had an invalid time bound and failed; a subsequent query succeeded.	The agent retrieved the active process-value summary from the platform.
Agent-proposed trial	The agent submitted a three-zone recipe trial from 198 to 200 °C. Manual approval preceded dispatch. The platform receipt and independent Modbus reads confirmed all three setpoints.	The source record links the task, approval, trial receipt, and register values. No separate response interval was defined for this step.
Follow-on approved step	A separate manually approved operation changed the three setpoints from 200 to 205 °C. Independent register reads and platform readback confirmed all three values.	Over the following 40 s, the simulated process value changed from 198.02 to 200.73 °C. Platform samples changed from 198.41 to 200.49 °C; the agent later retrieved 200.49 °C, matching the platform sample. Both actions occurred in one continuous run, so the observation does not isolate their contributions.

3 Interpretation and Limits

The test demonstrates one supervised path from production-context and acquisition queries through an agent-issued recipe trial, manual approval, Modbus device writes, independent register checks, and subsequent acquisition. The 2.71 °C change was measured in the 40 s window after the separate 200-to-205 °C step. The actions occurred in one continuous run without an isolated baseline between them, so their residual contributions cannot be separated.

The actions are not independent trials. The report does not provide a calibrated physical-process model, repeated runs, a matched controller, raw model-provider requests and responses, or a complete environment snapshot; the test therefore does not establish process-control benefit, model reasoning quality, or physical-plant performance.